Machine Learning Project Proposal

VoiceGuard: Local AI Scam Shield

Privacy-preserving real-time scam call detection and episodic memory companion for seniors

Part 1: Motivating Use Case

Protecting Seniors During Live Calls

- **Social Engineering Threat:** Scammers coerce seniors into sharing OTP codes.
- **One-Tap Speaker Shield:** Android overlay routes two-way audio locally.
- **100% Local Privacy:** Zero cloud dependency for financial or medical calls.
- **Episodic Memory Companion:** Auto-extracts clinic reminders from safe calls.



Part 1: Motivating Use Case

Financial Impact on Older Adults

3.2x

Higher average financial loss per senior victim compared to younger adults

Early detection within the first 20 seconds prevents sensitive financial disclosure.

Part 2: Problem Statement

Why Caller ID Blocking Fails

Traditional Number Blacklisting

- **Caller ID Spoofing:** Easy imitation of official bank numbers.
- **Rapid Number Rotation:** New VoIP lines spun up hourly.
- **Content Blindness:** Zero inspection of actual conversation dialogue.

AegisVoice ML Formulation

- **Acoustic Coupling:** Capturing two-way speakerphone audio at 16kHz PCM.
- **Cascading Inference:** Combining sub-50ms screening with deep reasoning.
- **Dual-Task Output:** Real-time threat levels plus post-call memory extraction.

Part 2: Problem Statement

Local Streaming Inference Pipeline



1. VAD & Whisper STT

3-second sliding window transcription.



2. Fast-Path ONNX

Sub-50ms DeBERTa scam screening.



3. Dual Qdrant RAG

BGE-M3 semantic threat matching.

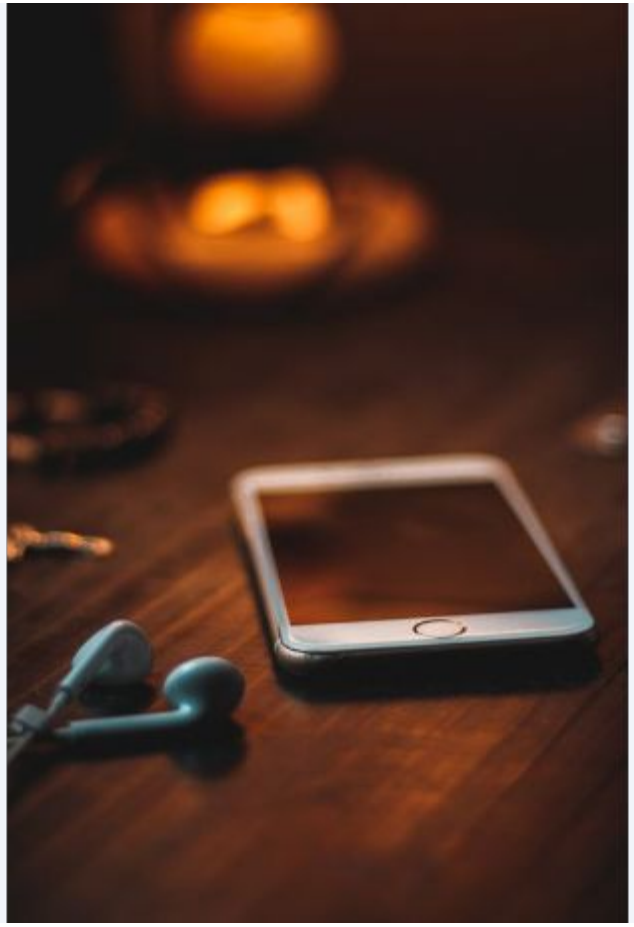


4. LLM & Voice Alert

Structured risk scoring plus TTS.

Part 3: Project Novelty

Hybrid Local AI & Episodic Memory



- **100% Local Inference:** Complete financial and medical privacy without cloud APIs.
- **Fast-Path + Vector RAG:** Combining ONNX classifiers with Qdrant threat similarity.
- **Episodic Memory Bank:** Indexing appointments from verified benign calls.
- **Dual-Interface Telemetry:** WCAG AAA senior shield plus caregiver command center.

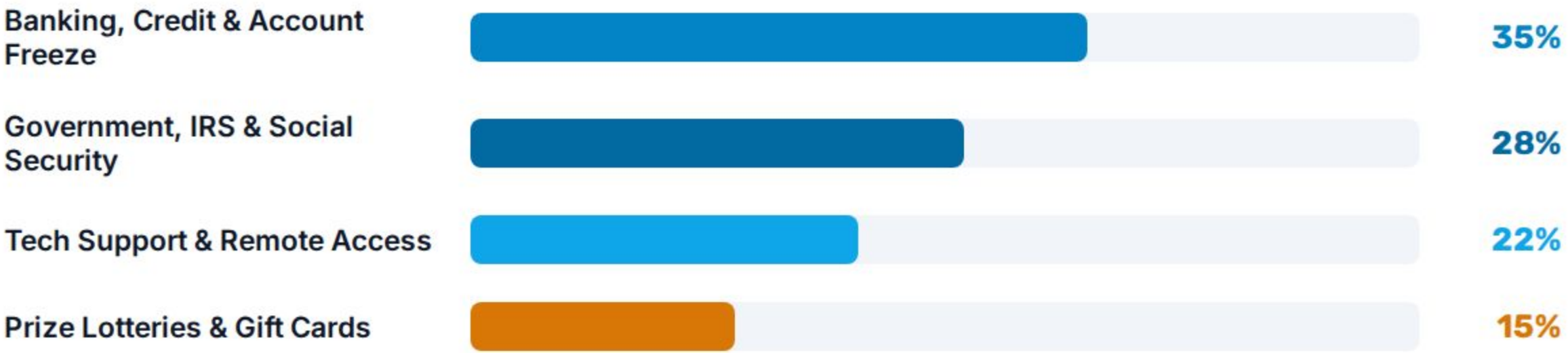
Part 4: Datasets

Publicly Available Datasets for Training & RAG

Dataset & Source	Modality & Format	Sample Size	Contribution to Model
BothBosu Scam Suite (HF)	Multi-turn Dialogues (CSV)	**5,800 calls**	Learning evolving scam progression
menaattia/phone-scam (HF)	Scam & Normal Transcripts (CSV)	**4,000 calls**	Impersonation vs. daily calls
Fraud_Call_Detection (GitHub)	Labeled Call Transcripts (TXT)	**5,925 calls**	Class imbalance resilience
ealvaradob/phishing (HF)	Phishing & Scam Corpus (Parquet)	**50,000+ records**	Fast-Path tuning & Qdrant RAG index
wspr-ncsu/robocall (GitHub)	Audio & Whisper Text (WAV)	**1,432 recordings**	Acoustic feature & STT evaluation
ivrit-ai/audio-v2 (HF)	Hebrew Speech & Text (WAV)	**20,000+ hours**	Benign calls & local adaptation

Part 4: Datasets

Scam Scenario Distribution Across Datasets



Part 5: Models & Architecture

Core AegisVoice ML Modules

Streaming Audio & STT

Faster-Whisper sliding window transcription.

Fast-Path & Vector RAG

ONNX DeBERTa plus Qdrant BGE-M3.

Local LLM & Memory

Ollama Qwen3 structured risk reasoning.

Cascading Model Progression

Stage 1

Fast-Path Classifier

Fine-tuned DeBERTa-v3 ONNX (<50ms)

Stage 2

Dual-Index Vector RAG

Qdrant hybrid search via BAAI/bge-m3Local

Stage 3

Deep Reasoning LLM

Qwen3-8B / Llama-3.1-8B JSON scorer

Part 6: Evaluation Plan

Success Metrics & Benchmarking



- **High Recall & F1-Score:** Minimizing missed real scam calls.
- **Tricky Benign Resilience:** Low false alarms on real clinic reminders.
- **p95 End-to-End Latency:** Measuring real-time streaming response speed.
- **Pipeline Ablation Study:** Fast-Path ONNX alone vs. full RAG-LLM hybrid.

Project Proposal Summary

AegisVoice: Local AI Scam Shield & Memory Companion

Thank you for listening — We welcome your questions and feedback